

SPINTRONICS ACT ONE

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Tutorial 1: The battery

Spintronic circuits are powered by a **battery**. The battery pulls chains through circuits, just like electrical batteries pull electrons through wires.

```
<div style="text-align: center;"></div>
```

Pull the string back and let it fly! It got going too fast, didn't it?

Good thing I built a circuit breaker into it!

Try it again. This time, after you pull back the string, **use your fingers to add some resistance to the output sprockets**. Can you release all of the energy in the battery without triggering the circuit breaker?

From now on, we'll have to avoid **shorts**! A short is wherever the battery has a path to release its energy without any resistance. How do you add resistance to the battery without using your fingers? With a resistor.

NEW PART UNLOCKED

```
<div style="text-align: center;"></div>  
<p style="text-align: center;"><b>THE SPINTRONIC RESISTOR</b></p>
```

PURPOSE:

RESISTORS RESIST THE FLOW OF CHAIN.

DETAIL:

RESISTANCE IS MEASURED IN "SPIN OHMS." THE HIGHER THE SPIN OHM VALUE OF A RESISTOR, THE HARDER IT IS TO TURN.

THE SYMBOL FOR A SPIN OHM IS "·"

Tutorial 2: Build a basic circuit

Let's build a basic circuit with a resistor and a battery. Before we can build a circuit with a resistor, we've got to make the workspace bigger.

Step 1. Place a tile in front of the battery.

```
<div style="text-align: center;">
```

Step 4. After you place a chain loop around sprockets, adjust the tension by sliding one part away from the other.

There should be just a tiny bit of slack in the chain!! If it's too tight, it will add extra resistance. If it's too loose, the chain will skip.

Step 5. Ok, we're ready! Pull the string and let go. Our first circuit!

HOW TO BUILD CHAINS

Start by making 8 separate chain loops of about 40 links each. Pieces snap either direction!

As you wrap the chain around sprockets to make circuits, add or remove links to shorten or lengthen the chain.

Use the blue links to help you see the circuit's speed and direction.

```
<div style="text-align: center;">
```

CHALLENGE 1: RESISTANCE

BUILD THE STARTING SETUP AND TEST THE CIRCUIT. THEN MODIFY THE CIRCUIT TO MAKE IT RUN SLOWER.

*Our first challenge! **How exciting!!***

*First, read the **challenge** on the wall.*

*Next, build the **starting setup**.*

*Your job is to figure out how to complete the challenge using the **available parts**.*

STARTING SETUP

```
<div style="text-align: center;">
```

AVAILABLE PARTS

- 1 × 1000 ·
- 1 × Chain

DIFFICULTY:

CHALLENGE 3: ONE CHAIN, TWO RESISTORS

MAKE BOTH RESISTORS TURN CLOCKWISE. YOU HAVE ONLY ONE CHAIN LOOP THIS TIME.

CLOCKWISE = ·

COUNTERCLOCKWISE = ·

STARTING SETUP

```
<div style="text-align: center;">
```

It's the same with spintronics. The spintronic battery doesn't **make** chain, it simply pulls a loop of chain around a circuit.

What happens if you don't make a loop?

In spintronics, the chain flies out onto the floor, and it eventually stops running. Luckily, in electronics, the electrons don't spill out onto the floor. That would leave a very dangerous situation in your house, and the power station would run out of electrons to push!

NEW PART UNLOCKED

```
<div style="text-align: center;"><b>THE SPINTRONIC SWITCH</b></p>
```

PURPOSE:

A SWITCH IS LIKE A MASSIVE RESISTOR THAT YOU CAN TOGGLE ON AND OFF.

DETAIL:

WHEN THE SWITCH IS ON (PUSHED DOWN), IT HAS NO RESISTANCE. WHEN THE SWITCH IS OFF (UP), IT HAS SUCH A HIGH RESISTANCE THAT YOU CAN'T EVEN TURN IT.

CHALLENGE 6: THE SWITCH

MAKE THE RESISTOR (R1) TURN ON AND OFF WITH A SWITCH.

```
<div style="text-align: center;">
```

Things that are analog could have an **infinite number of possible values** (like the graph on the left). The opposite of analog is "digital." Things that are digital can only have **certain values** (like the graph on the right).

Analog is pretty great, right?! The tricky part about things that are analog is that **every little imperfection** affects it. For example, the speed a spintronic resistor turns is affected by things like the temperature, air resistance, humidity, physical imperfections, and other things you can't easily control.

Eventually, you'll learn ways to design circuits that can **tolerate** imperfections, but for now, we're going to focus on one important thing that you can control: chain tension.

Connect two switches together with chain, like this:

```
<div style="text-align: center;">
```

STARTING SETUP

```
<div style="text-align: center;">
```

It's the same with electrons in wires. Signals travel fast through wires, but only because the wires are already full of electrons. Spintronic chains do this, too! If you pull one end of a long chain, the other end moves immediately, even though the chains aren't moving very fast.

```
<div style="text-align: center;">
```

The movement of chain (or electrons) through a circuit is called "current." A fast moving chain has high current. Current is measured in spin amps. The symbol for a spin amp is "A."

How fast is a spin amp? A chain traveling at 0.001 A moves this far each second: |-----|

NEW PART UNLOCKED

```
<div style="text-align: center;"><b>THE SPINTRONIC AMMETER</b></p>
```

PURPOSE:

THE AMMETER INDICATES THE AMOUNT OF CURRENT RUNNING THROUGH IT BY THE VOLUME AND PITCH OF THE SOUND IT MAKES.

IT'S MOST USEFUL WITH HIGHER CURRENTS THAT YOU CAN'T EASILY SEE WITH YOUR EYES.

CHALLENGE 9: MAKE SOME NOISE

MAKE THE AMMETER A1 TURN.

NOTE: THE AMMETER HAS VERY LITTLE RESISTANCE, SO MAKE SURE YOU USE A RESISTOR TO AVOID A SHORT!

STARTING SETUP

```
<div style="text-align: center;"><img
```


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AVAILABLE PARTS

- 1 × 1000 ·
- 1 × Chain
- 1 × 500 ·
- 1 × 200 ·

DIFFICULTY:

Tutorial 6: Circuit Diagrams

Ugh. I need a better way to draw these spintronic circuits. It's too much work to draw pictures of all the sprockets and chain.

Maybe I can use electronic circuit symbols. Let's see...

```
<div style="text-align: center;"></div>
```

...and the chains are just like wires! So a simple circuit like this:

```
<div style="text-align: center;"></div>
```

That's WAY easier! And we'll say the '+' symbol on the battery is where it's <u>pushing</u> chain and the '-' is where it's <u>pulling</u> chain.

Wait a second. What about these? How can we draw circuit diagrams for them?

```
<div style="text-align: center;"></div>
```

They're all the same circuit! In each case, the resistors and battery are directly connected together. In other words, you can't turn one without turning the others! <u>Any parts that are connected directly together must be drawn in the same loop.</u> Like this:

```
<div style="text-align: center;"></div>
```

NOTE: REMEMBER TO CHECK YOUR SOLUTION IN THE BACK OF THE BOOK WHEN YOU SEE THIS .
THERE ARE MANY EQUIVALENT SOLUTIONS.

Time to try a few **circuit diagram puzzles!** I have to remember these **symbols**:

```
<div style="text-align: center;"></div>
```

STARTING SETUP

```
<div style="text-align: center;">
```

EXTRA CREDIT: BUILD THE SAME CIRCUIT TWO MORE WAYS.

.

STARTING SETUP

```
<div style="text-align: center;">` symbol represents the '+' side of the battery and the `` symbol represents "ground." Ground is usually the place where chain returns to the battery at the '-' side.

## CHALLENGE 15: SIMPLIFIED CIRCUIT DIAGRAM

BUILD THE FOLLOWING CIRCUIT:

```
<div style="text-align: center;">
.
```

STARTING SETUP

```
<div style="text-align: center;">
```

.

*The little bump shows that the wires pass by each other here, but **aren't connected**.*

STARTING SETUP

```
<div style="text-align: center;">THE SPINTRONIC JUNCTION</p>
```

PURPOSE:

THE JUNCTION ALLOWS CURRENT IN ONE LOOP TO SPLIT INTO TWO LOOPS.

DETAIL:

THIS IS JUST LIKE AN ELECTRONIC JUNCTION, WHERE ONE WIRE SPLITS INTO TWO. IT'S AN EXTREMELY USEFUL PART, BUT VERY DIFFICULT TO MASTER.

## Tutorial 9: Spintronic junctions

Spintronic junctions allow current in one loop to split into two loops. Think of it like a junction between three pipes:

```
<div style="text-align: center;"></div>
```

It's the same with a spintronic junction.

```
<div style="text-align: center;"></div>
```

You can push current into **any** of the three sprockets and it will split between the other two. Junctions are mind-bending, but they make it possible to build just about any electronic circuit.

## CHALLENGE 17: FIX THE SHORT CIRCUIT

Fix the short.

NOTE: WATCH OUT FOR SHORTS WHEN YOU USE JUNCTIONS. It's EASY TO ACCIDENTALLY MAKE THEM.

The number of little gray dots on the chains show the sprocket they're meant to be on:

```
<div style="text-align: center;"></div>
```

#### STARTING SETUP

```
<div style="text-align: center;">
<table>
 <tr>
 <th colspan="2">INPUT</th>
 <th>OUTPUT</th>
 </tr>
 <tr>
```

```

 <th>S1</th>
 <th>S2</th>
 <th>A1</th>
 </tr>
 <tr>
 <td>OFF</td>
 <td>OFF</td>
 <td>NO SOUND</td>
 </tr>
 <tr>
 <td>OFF</td>
 <td>ON</td>
 <td>SOUND</td>
 </tr>
 <tr>
 <td>ON</td>
 <td>OFF</td>
 <td>SOUND</td>
 </tr>
 <tr>
 <td>ON</td>
 <td>ON</td>
 <td>SOUND</td>
 </tr>
</table></div>

```

I need a circuit that **behaves like this table**.

Each row shows a combination of S1 and S2, and whether A1 should make sound.

We need to make one circuit that works for **all four cases**.

## STARTING SETUP

```

<div style="text-align: center;"></div>

```

## AVAILABLE PARTS

- 1 × 200 ·
- 4 \time Chain
- 1 × Junction

DIFFICULTY: .....

## Tutorial 10: Parallel circuits

Are you starting to get the hang of junctions? Before junctions, everything was connected in **series**: If one part turns, the other parts turn. But when you put a junction between parts, they are connected in "**parallel**": turning one part doesn't necessarily make other parts turn.

Here's a series circuit and a parallel circuit with two resistors:

```

<div style="text-align: center;"></div>
```

It's called "parallel" because the junction splits the current between two circuits that run independently, or "in parallel", with each other.

**Try it!** Build each circuit. If you stop one resistor with your fingers, what happens to the other resistor?

## CHALLENGE 20: WHERE IS THE HIGHEST CURRENT?

BUILD THE CIRCUIT BELOW. THEN ADD THE AMMETER AT THE LOCATION WITH THE HIGHEST CURRENT (THAT IS, THE PLACE THAT MAKES IT PRODUCE THE HIGHEST PITCH POSSIBLE).

.

### STARTING SETUP

```
<div style="text-align: center;"></div>
```

#### AVAILABLE PARTS

- 1 × 500 ·
- 2 × Chain
- 1 × Junction

DIFFICULTY: .....

### CHALLENGE 23: PITCH SWITCH

WHEN S1 IS ON, A1 MUST MAKE A HIGHER PITCH SOUND THAN WHEN S1 IS OFF.

```
<div style="text-align: center;"></div>
```

.

#### STARTING SETUP

```
<div style="text-align: center;"></div>
```

#### AVAILABLE PARTS

- 1 × 500 ·
- 4 × Chain
- 1 × 200 ·
- 1 × Junction

DIFFICULTY: .....

### CHALLENGE 24: CHANGING LEVELS

COMPLETE THE CIRCUIT BY CONNECTING JUNCTION J1 TO JUNCTION J2.

**NOTE:** THE SWITCH SHOULD ALWAYS STAY IN THE ON POSITION IN THIS CIRCUIT.

**Hold up a second!** How can I connect J1 to J2 when the only free sprockets are on **different levels**?

#### STARTING SETUP

```
<div style="text-align: center;"></div>
```



## AVAILABLE PARTS

- 2 × Chain
- 1 × Switch

DIFFICULTY: .....

## Tutorial 11: Parallel Circuit Diagrams

We've been using junctions to build a lot of parallel circuits, but how do we draw them in a circuit diagram? Here's the most basic parallel circuit...

```
<div style="text-align: center;">
```

The junction splits the green loop into **two parallel paths**: a yellow path and a blue path. Those two paths each go through a resistor and then rejoin the green loop again.

So how would a simplified circuit diagram of that same circuit look? Like this...

```
<div style="text-align: center;">
```

As before, we can skip drawing the part where the paths connect back to the negative side of the battery. Let's try some more...

## CHALLENGE 25: PARALLEL AND SERIES TOGETHER

BUILD THE FOLLOWING CIRCUIT:

```
<div style="text-align: center;"></div>
```

.

*Don't we need a junction with 4 sprockets to make this?*

*Wait, no! We can do it! It's just the same circuit as this:*

```
<div style="text-align: center;"></div>
```

STARTING SETUP

```
<div style="text-align: center;"></div>
```

AVAILABLE PARTS

- 1 × 1000 ·
- 5 × Chain
- 1 × 500 ·
- 1 × 200 ·

DIFFICULTY: .....

## CHALLENGE 27: PARALLEL AND SERIES 2

BUILD THE FOLLOWING CIRCUIT:

```
<div style="text-align: center;"></div>
```

.

STARTING SETUP

```
<div style="text-align: center;"></div>
```

AVAILABLE PARTS

- 1 × 1000 ·
- 4 × Chain
- 1 × 500 ·
- 1 × Switch
- 1 × Junction

DIFFICULTY: .....

## CHALLENGE 28: PARALLEL AND SERIES 3

BUILD THE FOLLOWING CIRCUIT:

```
<div style="text-align: center;"></div>
.
```

STARTING SETUP

```
<div style="text-align: center;"></div>
```

AVAILABLE PARTS

- 8 × Chain

DIFFICULTY: .....

## CHALLENGE 29: PARALLEL AND SERIES 4

BUILD THE FOLLOWING CIRCUIT:

```
<div style="text-align: center;"></div>
.
```

STARTING SETUP

```
<div style="text-align: center;"></div>
```

*This one's **tricky**. Of course I could make it with two junctions, but I can also make it with **just one**!*

*Think of this section of the circuit as **one unit** that's in series with the 200 · resistor. In the unit, one loop of the junction goes to the 500 · resistor and back. Another loop goes to the 1000 · resistor and back.*

```
<div style="text-align: center;"></div>
```

AVAILABLE PARTS

- 3 × Chain

DIFFICULTY: .....

## CHALLENGE 30: PARALLEL AND SERIES 5

BUILD THE FOLLOWING CIRCUIT:

```
<div style="text-align: center;">
.
```

STARTING SETUP

```
<div style="text-align: center;">THE SPINTRONIC CAPACITOR</p>
```

PURPOSE:

A CAPACITOR STORES ENERGY. TURN IT TO STORE ENERGY, THEN RELEASE IT.

DETAIL:

CAPACITANCE IS MEASURED IN "SPIN FARADS." THE SYMBOL FOR SPIN FARAD IS 'F'.

## CHALLENGE 31: CHARGE THE CAPACITOR

MAKE THE BATTERY CHARGE THE CAPACITOR.

**NOTE:** TURN THE CAPACITOR SO BOTH HANDS POINT TO ZERO BEFORE YOU CHARGE THE BATTERY.

STARTING SETUP

```
<div style="text-align: center;"></div>
```

*The first time you turn on the switch, the **capacitor charges**.*

*After that, it just...**stays charged**.*

STARTING SETUP

```
<div style="text-align: center;"></div>
```

STARTING SETUP

```
<div style="text-align: center;"></div>
```

STARTING SETUP

```
<div style="text-align: center;"></div>
```

***Yikes!** Just don't turn **both** switches on at the same time!*

STARTING SETUP

```
<div style="text-align: center;"></div>
```

*When the capacitor discharges through a resistor, you might notice that it doesn't discharge **ALL the way** to zero.*

*That's ok - it's an imperfection in the system that I couldn't completely work out.*

```
<div style="text-align: center;"></div>
```

.

*Hmm...how can I charge and discharge the capacitor with only **one** button?*

#### STARTING SETUP

```
<div style="text-align: center;"></div>
```

#### STARTING SETUP

```
<div style="text-align: center;"><img

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```

#### AVAILABLE PARTS

- 1 × 500 ·
- 5 × Chain
- 1 × 200 ·
- 1 × Capacitor (0.001 F)
- 1 × Switch
- 1 × Junction

DIFFICULTY: .....

### CHALLENGE 41: CAPACITOR BACKUP

BUILD A CIRCUIT THAT CONTINUES TURNING A1 EVEN WHEN THE POWER IS SHUT OFF FOR A SHORT WHILE BY S1.



```
<div style="text-align: center;">
```

I need the power to **keep running** even when the battery power **goes out** for a little while. This ought to help.

## STARTING SETUP

```
<div style="text-align: center;">
```

## AVAILABLE PARTS

- 3 × Chain
- 1 × Capacitor (0.001 F)
- 1 × Junction

DIFFICULTY: .....

## Tutorial 12: Voltage

Voltage is a measure of how hard the chain pulls on a part in the circuit. Voltage is measured in "spin volts", or "V". Spintronic capacitors can be used to measure the voltage. In the picture below, a sprocket pushes on the capacitor with +6 V.

```
<div style="text-align: center;">
```

And here, the sprocket doesn't push at all. The spintronic capacitor shows 0 V.

```
<div style="text-align: center;"></div>
```

STARTING SETUP

```
<div style="text-align: center;">
```

The battery produces 6 V, and it pushes chain through a loop with three resistors. Each of those resistors push back against the current. That "push back" is also a voltage - a **negative voltage**! Bigger resistors generate more negative voltage than smaller resistors. Add all the voltages in the loop together and what do you get? Zero spin volts!

There's a term for the negative voltage generated by each resistor. It's called the "**voltage drop**". That's because the voltage drops by that amount from one side of the resistor to the other. In the next picture, I drew in the voltage at each point in the circuit.

```
<div style="text-align: center;"></div>
```

At first, this was confusing for me. Doesn't the capacitor create another path for current? It doesn't really. The capacitor charges up quickly and then blocks any more current from flowing through that path. It just charges to the **difference in voltage** from point A to point B. That is, it charges to **the voltage across the 200 · resistor**

This still might be hard to understand, so try this. Build the circuit above, but take out the capacitor, like this:

```
<div style="text-align: center;"></div>
```

.

*I need to measure the voltage across R2, so I'll need to place a capacitor (voltmeter) in parallel with it.*

### STARTING SETUP

```
<div style="text-align: center;">
```

*The output is the opposite of the input? This could be useful...*

### STARTING SETUP

```
<div style="text-align: center;">
```

### AVAILABLE PARTS

- $1 \times 500\ \Omega$
- $4 \times$  Chain
- $1 \times$  Junction

DIFFICULTY: .....

## CHALLENGE 49: NAND GATE

THE VOLTMETER (CAPACITOR) C1 MUST SHOW 0 V IF S1 AND S2 ARE ON. OTHERWISE IT MUST SHOW ABOUT 6 V. (NEGATIVE 6 V IS ALSO ACCEPTABLE.)

```
<div align="center">
<table>
 <tr>
 <th colspan="2">INPUT</th>
 <th>OUTPUT</th>
 </tr>
 <tr>
 <th>S1</th>
 <th>S2</th>
 <th>C1</th>
 </tr>
 <tr>
 <td>OFF</td>
 <td>OFF</td>
 <td>6V</td>
 </tr>
 <tr>
 <td>OFF</td>
 <td>ON</td>
 <td>6V</td>
 </tr>
 <tr>
 <td>ON</td>
 <td>OFF</td>
 <td>6V</td>
 </tr>
 <tr>
 <td>ON</td>
 <td>ON</td>
 <td>0V</td>
 </tr>
</table></div>
```

### STARTING SETUP

```
<div style="text-align: center;">
<table>
 <tr>
 <th colspan="2">INPUT</th>
 <th>OUTPUT</th>
 </tr>
 <tr>
 <th>S1</th>
 <th>S2</th>
 <th>C1</th>
 </tr>
 <tr>
 <td>OFF</td>
 <td>OFF</td>
 <td>6V</td>
 </tr>
 <tr>
 <td>OFF</td>
 <td>ON</td>
 <td>0V</td>
 </tr>
 <tr>
 <td>ON</td>
 <td>OFF</td>
 <td>0V</td>
 </tr>
 <tr>
 <td>ON</td>
 <td>ON</td>
 <td>0V</td>
 </tr>
</table></div>
```

#### STARTING SETUP

```
<div style="text-align: center;"></div>
```

#### AVAILABLE PARTS

- 1  $\times$  500 Ohm
- 6  $\times$  Chain
- 2  $\times$  Junction

DIFFICULTY: .....

## CHALLENGE 51: PITCH SWITCH REVERSED

WHEN THE SWITCH S1 IS OFF, A1 SHOULD TURN FASTER THAN WHEN S1 IS ON.



```
<div style="text-align: center;"></div>
```

**NOTE:** THIS CHALLENGE IS HARD. FEEL FREE TO SKIP IT AND COME BACK AGAIN LATER.

(S)

```
<div style="text-align: center;">
```

First, look at the picture and find the sprocket you are applying a voltage to (like with the battery). Then follow the arrows to the other sprockets. If it's a green arrow, it will push that sprocket with the **same** voltage. If it's a red arrow, it will push it with the **opposite** voltage.

For example, if you apply a voltage to the middle sprocket, it pushes on the bottom sprocket in the **same** direction, but it pushes on the top sprocket in the **opposite** direction. Or if you apply a voltage to the bottom sprocket, it pushes on the middle sprocket in the same direction and the top sprocket in the **same** direction.

## CHALLENGE 52: JUNCTION SPIN DIRECTION

R1 MUST TURN COUNTERCLOCKWISE AND R2 MUST TURN CLOCKWISE.

STARTING SETUP

```
<div style="text-align: center;"></div>
```

Let's mark the branch that's attached to the bottom sprocket of the junction with a little dot, like this:

```
<div style="text-align: center;"></div>
```

Now, when you're trying to figure out which way a voltage will push on a branch, take your finger and trace the path. **If it passes a dot, the voltage will push the <u>same</u> direction through the branch. Otherwise, it will push the <u>opposite</u> direction.**

For example, in this circuit, start by tracing the branch to the right. It passes a dot, so the voltage pushes in the same direction.

```
<div style="text-align: center;"></div>
```

Now trace the branch to the left. It does not pass a dot, so the voltage pushes in the opposite direction.

```
<div style="text-align: center;"></div>
Here's the same basic circuit, but this time the battery is
connected to the bottom sprocket. Now both branches pass a dot, so
the voltage pushes in the same direction for both.
<div style="text-align: center;"></div>
Notice I didn't draw a dot where the branches come back together. I
could, but it's unnecessary.
```

Let's practice...

## CHALLENGE 55: JUNCTION DOT DIAGRAMS 1

BUILD THE FOLLOWING CIRCUIT:

```
<div style="text-align: center;"></div>
```

(S)

STARTING SETUP

```
<div style="text-align: center;"></div>
```

#### AVAILABLE PARTS

- 5 \times Chain

DIFFICULTY: .....

### CHALLENGE 56: JUNCTION DOT DIAGRAMS 2

BUILD THE FOLLOWING CIRCUIT:

```
<div style="text-align: center;"></div>
IF CORRECT:
<div style="text-align: center;"></div>
(S)
```

#### STARTING SETUP

```
<div style="text-align: center;"></div>
```

#### AVAILABLE PARTS

- 5 \times Chain

DIFFICULTY: .....

### CHALLENGE 57: ALL THE SAME DIRECTION

BUILD A CIRCUIT WHERE ALL THREE RESISTORS ARE IN PARALLEL AND THEY ALL TURN CLOCKWISE.

*A switch?!? Oh yeah! Sometimes I need a switch left in the ON position to change levels.*

#### STARTING SETUP

```
<div style="text-align: center;"></div>
```

#### AVAILABLE PARTS

- 6 \times Chain
- 1 \times Switch

DIFFICULTY: .....

## Tutorial 18: Spintronic Coupling

Most spintronic parts have three sprockets that are joined together. Take the resistor, for instance. All three sprockets are joined, so that if one moves, the other two also move. We say that **chains attached to joined sprockets are "coupled" together.**

```
<div style="text-align: center;">
```

The top chain loop turns because it's "coupled" to the bottom chain loop at the 1000 Ohm resistor. That dashed line between the 1000 Ohm resistor and the top loop tells us that **both loops wrap around that resistor**. Of course, that means the current in each loop **must be the same**.

Now, do you see those two dots next to the coupling? Since they're both on the **same end** of the coupling, they tell us that both chains turn the **same direction** around the resistor (clockwise).

Now you're probably thinking, "Are you crazy? Why not just draw it like the first diagram - it's way simpler!" Well, yeah, in this case, the first diagram is much simpler, and you can just draw it that way, but sometimes you have to draw the coupling.

For instance, here the two loops are **coupled oppositely**. The positive current through the resistor of the bottom loop becomes a negative current through the top loop.

```
<div style="text-align: center;">
```

The corresponding circuit diagram looks like this:

```
<div style="text-align: center;"></div>

There's only one part you can't directly couple to: the junction. That's because it doesn't have any joined sprockets at all. Still, if you need to, you can always connect the junction to another part (like a switch) and couple to that part instead.

Enough talk about coupling, **let's try it out!**

CHALLENGE 58: COUPLING DIAGRAMS 1

BUILD THE FOLLOWING CIRCUIT:

```
<div style="text-align: center;"></div>  
(S)
```

STARTING CIRCUIT

```
<div style="text-align: center;"></div>  
(S)
```

STARTING SETUP

```
<div style="text-align: center;"></div>
```

Wow, that switch couples **three different loops!**

STARTING SETUP

```
<div style="text-align: center;"></div>
```

Tricky! It looks like **two loops** from the junction will be connected to R1.

Interesting. Both loops are going to push on R1, and since **both of them push with 6 V**, the total push is...

STARTING SETUP

```
<div style="text-align: center;">
```

AVAILABLE PARTS

- 6 \times Chain
- 2 \times Switch
- 2 \times Junction

DIFFICULTY:

CHALLENGE 65: LOCKED CIRCUIT 2

BUILD A CIRCUIT THAT CAN'T MOVE AT ALL.

STARTING SETUP

```
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```

AVAILABLE PARTS

- 1 $\times$ 500 Ohm
- 7 $\times$ Chain
- 1 $\times$ 200 Ohm
- 2 $\times$ Junction

DIFFICULTY:

END OF ACT ONE

SELECTED SOLUTIONS

This section contains solutions for selected puzzles. Hints and solutions for all of the challenges can be found online at www.spintronics.com/solutions. Remember that these solutions are often just one of *many* possibilities.

CHALLENGE 12

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CHALLENGE 13

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CHALLENGE 14

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CHALLENGE 15

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CHALLENGE 16

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CHALLENGE 20

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CHALLENGE 21

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CHALLENGE 23

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CHALLENGE 25

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CHALLENGE 26

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CHALLENGE 27

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CHALLENGE 28

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CHALLENGE 29

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CHALLENGE 30

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CHALLENGE 39

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CHALLENGE 45

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CHALLENGE 47

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CHALLENGE 51

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CHALLENGE 55

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CHALLENGE 56

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CHALLENGE 58

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CHALLENGE 59

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CHALLENGE 60

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CHALLENGE 61

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CHALLENGE 62

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CHALLENGE 64

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CHALLENGE 66

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CHALLENGE 67

```
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```